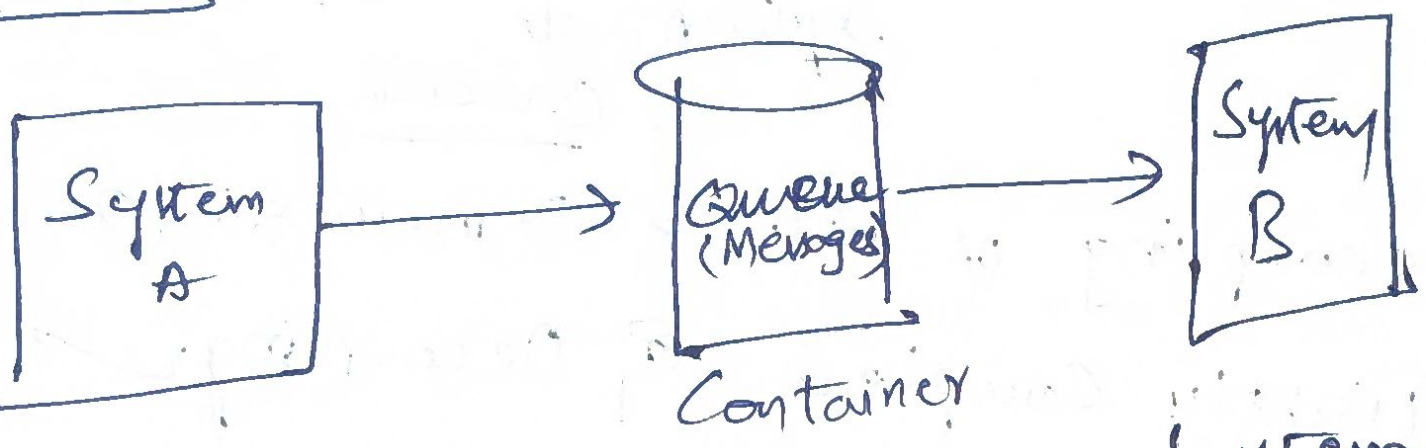
Day 51 JMS queue (Java Messaging Service)

* It's very Important
* IBMMQ, Kafka, etc Messaging queue Servers
* Messaging Queues → Somebody will publish the message to that queue and → Somebody will utilize that message
* Example:



* No direct Interaction between System A and System B for message exchanging process
* Event driven architecture also Same as JMS
* So here JMS → used from SAP CPI we don't use Third party Queues → Instead we will use JMS Queues (Internally) in SAP BTP CPI/IS

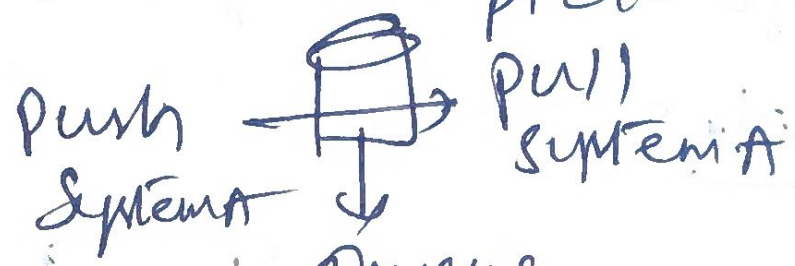
→ JMS Queue → Retry machanism, Data persistance

↓

Mostly we can use it

→ Scenario: iflow1 → message Place → Queue

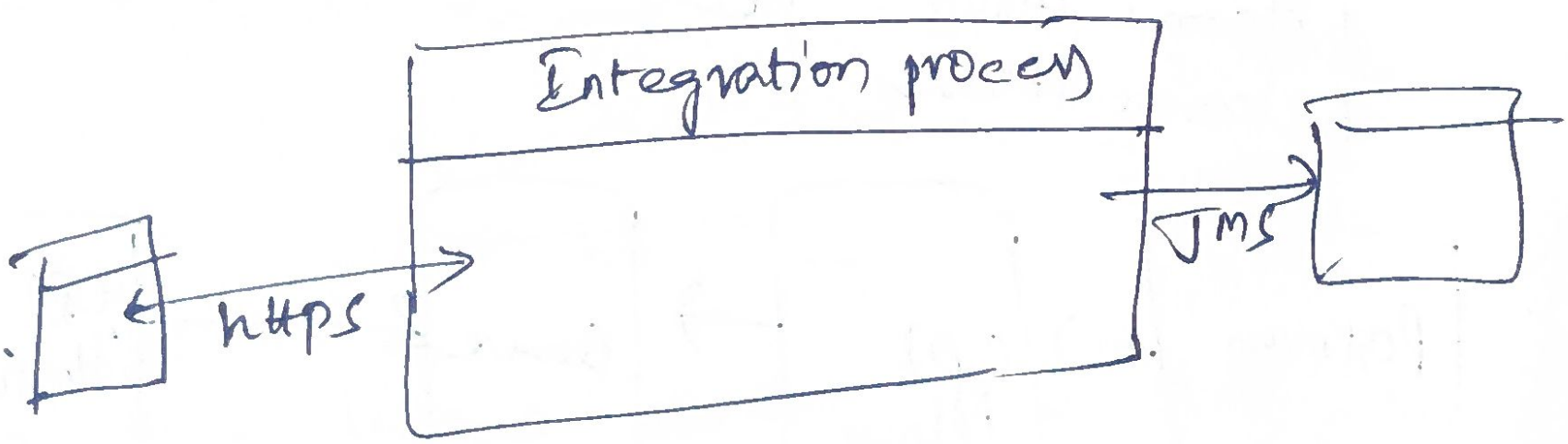
→ iflow2 → message pick



* Decoupling of iflow's → messaging Queues.
* Tightly Coupled & Decoupling.
* Retry mechanism & Decoupling we will use JMS Queues
* Retry Machanism → Suppose Interface failed It will be reprocessed again (JMS Queue)
* all message Queues are Same process
* If we want to use other message Queue from Sap api to third party Messaging Queues the we will use AMQP Adapter

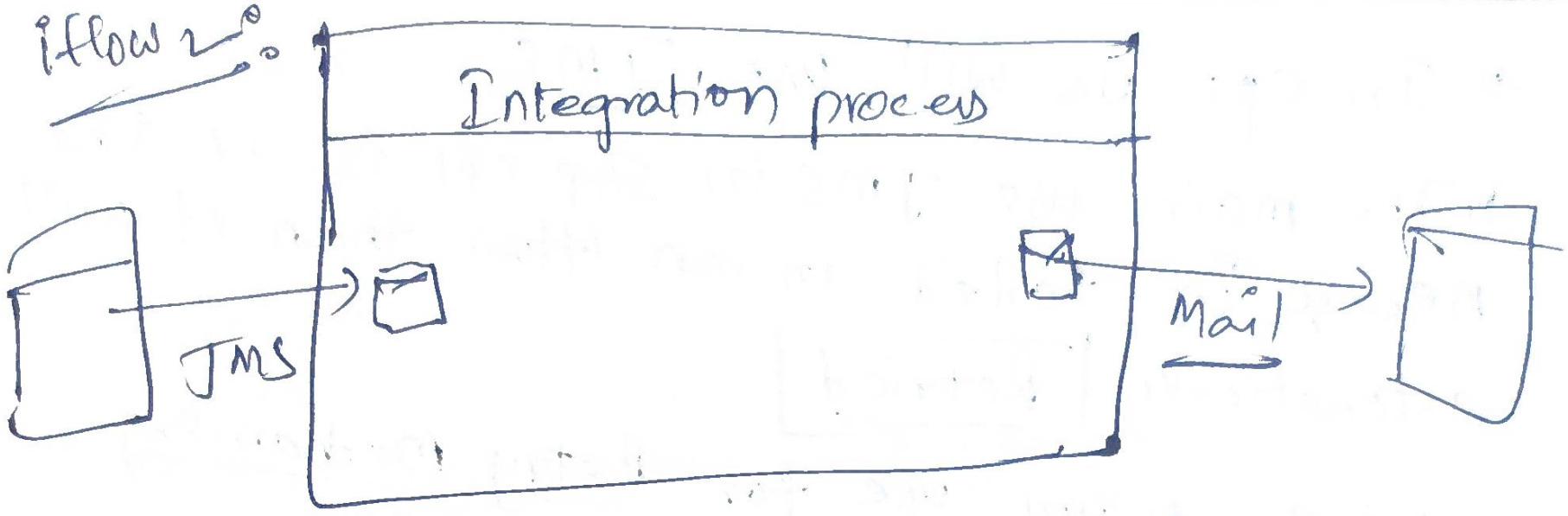
* In CPI we will use JMS
* The main use JMS in SAP CPI is if the message is failed in an iflow then it will automatically Retried
* What do you use for Retry Machanism in SAP CPI is JMS queues
* Guarantie & message transfer → JMS queues
→ popSub model
* Demo: CPI doesn't support Retry mechanism automatically ~ So we need to use JMS queues only

⇒ Scenario: iflow1 → Push the message to JMS Queue



iflow2 consume that message from the iflow)

iflow 2:



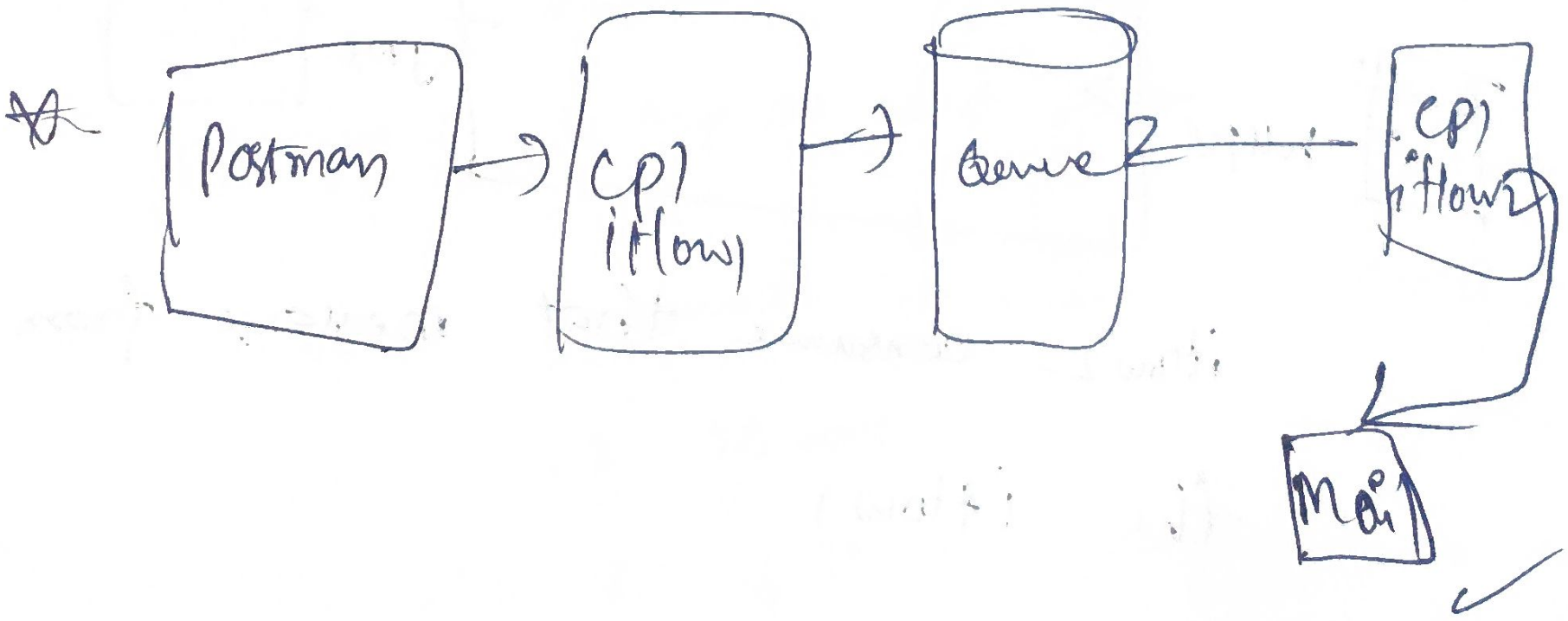
* JMS queues → manage stores → message queues

iflow1 (write)
iflow2 (consume)

* https → mail → fail
(server down)

* But in JMS queue → ?

→ Hi Pavan
Welcome iflow1
Welcome iflow2



* Message queue → is always Asynchronous Message

one way communication ✓

Don't wait for Response

* always → queue's → Asynchronous ✓

→ Wrong mail Server → Testing

⇒ if flow1 Success, But if flow2

* How will you do retry in CPI ✓ (Main Question), Error handling, reprocessing failed Messages, guarantee delivery how you will give ✓

Source → Target

* JMS Queue's → untill or unless it the Message is Successful → the message will be in that Queue

* Now discuss about the JMS adapter

* first General tab

* processing Tab:

Queue Name: Random name

Access Type: Exclusive, Non exclusive

Retention Threshold for Alerting (in'd)* 2 (Due at monitor tab)

Expiration period (in'd)* 30

compress Stored message [✓]

Encrypt Stored message [ ]

Transfer Exchange properties:

Inside CPI

eg: Access Type → 10 msgs → worker nodes.

parallel processing, and sequentional processing

(non-Exclusive) (one by one)

(Exclusive)

* Now JMS Sender Adapter

Connection Tab:

Queue Name: Random name.

Access type: Non exclusive, exclusive

Number Of Concurrent procs: 1 (NO for exclusive)

Retry Details:

Retry Interval (in min) * 1

Exponential Back off: ☑

Maximum Retry interval (in min): 60

Dead-letter Queue: ☑

→ It will double the retry Interval

5, 10, 20, 40, 80 → min

→ It will works only with the Acces Type → non-Exclusive only

↓

parallel processing

Out of memory

* Any msg Queue → Suppose → msg → maximum retry → Dead letter queue

JMS queue → msg → out of memory → worker Node → problem → msg → dead letter queue

ie

Only out of memory issue → msg → Dead letter queue (After 2 time's retry)

* Realtime → Unlimited Retry's → not Recommended

* Flows

